

hyper2kvm — Ubuntu 22.04 LVM vCenter Migration Report

hyper2kvm Migration Team

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Ubuntu 22.04 LVM — vCenter to KVM Migration Report

Date: March 19, 2026 **Tool:** hyper2kvm v0.2.2 **Source:** VMware vCenter 10.73.213.134 (ESXi 8.0) **Target:** Fedora 43 (KVM/libvirt)

1. Source VM Details

Parameter	Value
VM Name	esx8.0-ubuntu22.04.5-x64-with-lvm-partitions
vCenter	10.73.213.134
Datacenter	data
ESXi Host	10.73.212.36
Guest OS	Debian GNU/Linux 12 (64-bit)
Actual OS	Ubuntu 22.04.5 LTS
Memory	2048 MB
vCPUs	1
Disk	LVM partitions
Power State	poweredOff
Firmware	UEFI

2. Migration Config

File: ubuntu-vcenter-to-libvirt.yaml

```
cmd: vsphere
vs_action: export_vm
export_mode: ovf_export
vs_control_plane: govc

vcenter: 10.73.213.134
vc_user: administrator@vsphere.local
vc_password: VCENTER@redhat2025
vc_insecure: true
dc_name: data
govc_url: "https://10.73.213.134/sdk"
govc_insecure: true
govc_datacenter: data
govc_export_remove_cdroms: true

vm_name: esx8.0-ubuntu22.04.5-x64-with-lvm-partitions

output_dir: ./output-ubuntu
flatten: true
to_output: ubuntu-2204.qcow2
out_format: qcow2
compress: true

fstab_mode: stabilize-all
regen_initramfs: true
remove_vmware_tools: true

emit_domain_xml: true
virsh_define: true
memory: 2048
vcpus: 2
uefi: true
machine: q35
disk_bus: virtio
disk_cache: none
net_model: virtio
libvirt_network: default
serial_console: true
graphics: vnc
guest_os: linux
```

```
libvirt_test: true
keep_domain: true
timeout: 120
```

```
verbose: 1
```

Command:

```
sudo ./h2kvmctl --config ubuntu-vcenter-to-libvirt.yaml
```

3. Migration Pipeline

```
vCenter (10.73.213.134 / datacenter: data)
|
|   govc export.ovf (NFC streaming, ~3 hours)
|   ▼
OVF + VMDK (5.4 GB)
|
|   Auto-chained (no manual step)
|   ▼
VMCraft offline fixes
|   ├── fstab stabilized (UUID-based mounts)
|   ├── initramfs rebuilt (virtio_blk, virtio_scsi, virtio_net,
nvme)
|   ├── LVM volumes activated and fixed
|   └── VMware Tools removed
|
|   qemu-img convert (flatten + compress)
|   ▼
ubuntu-2204.qcow2 (5.4 GB compressed)
|
|   emit libvirt domain XML
|   ▼
libvirt domain defined + started
|
|   boot test
|   ▼
VM RUNNING on KVM ✓
```

4. Results

Export Phase

Metric	Value
Export method	govc export.ovf (NFC)
Export time	~3 hours
VMDK size	5.4 GB
OVF size	9.3 KB
CD-ROM removed	cdrom-3000

Conversion Phase

Metric	Value
Backend	VMCraft (pure Python + qemu-nbd)
Input	5.4 GB VMDK
Output	ubuntu-2204.qcow2 (5.4 GB compressed)
Format	qcow2
Pipeline chaining	Automatic (govc → convert → fixes → libvirt)

Guest Fixes Applied

Fix	Status
fstab stabilization	Applied (UUID-based)
initramfs rebuild	Applied (virtio_blk, virtio_scsi, virtio_net, nvme)
LVM activation	Activated
VMware Tools removal	Removed
Boot configuration	UEFI (systemd-boot)

libvirt Deployment

Metric	Value
Domain name	esx8.0-ubuntu22.04.5-x64-with-lvm-partitions
Domain XML	

Metric	Value
	output-ubuntu/libvirt/esx8.0-ubuntu22.04.5-x64-with-lvm-partitions.xml
Machine type	q35
Firmware	UEFI (OVMF)
Disk bus	virtio
Network	virtio (default network)
Graphics	VNC
virsh define	SUCCESS
virsh start	SUCCESS
Boot test	PASSED — domain reached RUNNING state

5. Output Files

```
output-ubuntu/
├─ ubuntu-2204.qcow2
(5.4 GB)
├─ libvirt/
|   └─ esx8.0-ubuntu22.04.5-x64-with-lvm-partitions.xml
(1.9 KB)
└─ esx8.0-ubuntu22.04.5-x64-with-lvm-partitions.VARS.fd
(NVRAM)
└─
work/
(temp)
```

6. Verification

```
$ sudo virsh list | grep ubuntu
20    esx8.0-ubuntu22.04.5-x64-with-lvm-partitions    running

$ ls -lh output-ubuntu/ubuntu-2204.qcow2
-rw-r--r-- 1 qemu qemu 5.4G output-ubuntu/ubuntu-2204.qcow2

$ ls -lh output-ubuntu/libvirt/*.xml
```

```
-rw-r--r-- 1 root root 1.9K output-ubuntu/libvirt/esx8.0-ubuntu22.04.5-x64-with-lvm-partitions.xml
```

7. Target Host Environment

Component	Version
OS	Fedora 43 (kernel 6.18.13)
hyper2kvm	0.2.2
govc	0.44.0
qemu-img	10.1.4
libvirt	11.6.0
OVMF	OVMF_CODE.fd
Backend	VMCraft (pure Python)

8. Conclusion

Result: SUCCESS

Ubuntu 22.04.5 LTS with LVM partitions was successfully migrated from VMware vCenter (ESXi 8.0) to KVM/libvirt using a single `h2kvmctl --config` command. The entire pipeline — govc NFC export, disk conversion, offline guest fixes (fstab, initramfs, LVM, VMware Tools), libvirt domain creation, and boot test — ran automatically without manual intervention.